



de Broglie Wavelength

Key Objectives

- On the atomic level, energy and matter exhibit the characteristics of both waves and particles. Young's double slit experiment is explained with a wave model but produces the same interference and diffraction patterns when one photon at a time or one electron at a time are passed through the slits

Relevant Equations

$$\textit{de broglie } \lambda = \frac{h}{p}$$

de Broglie wavelength

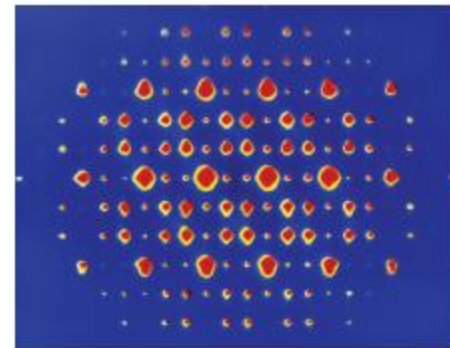
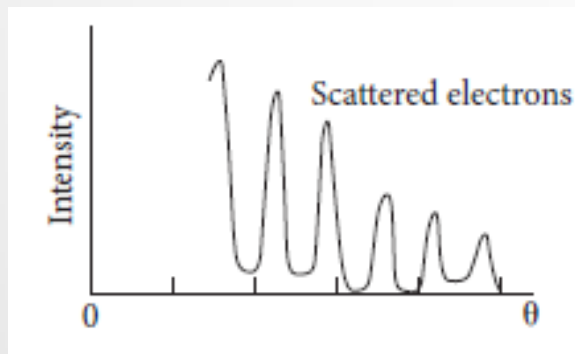
- Combines momentum equation $p = mv$ with the equation for the momentum of light $p = \frac{h}{\lambda}$.

$$\lambda = \frac{h}{mv}$$

- Concluded that if light can behave like a particle, then matter can behave like a wave.
- Evidence to support this came later, the Davisson-Germer experiment observed electron diffraction which could only be explained if particles have a wave like nature.

Electron diffraction

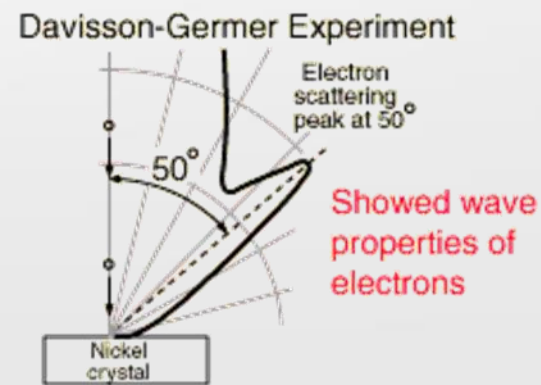
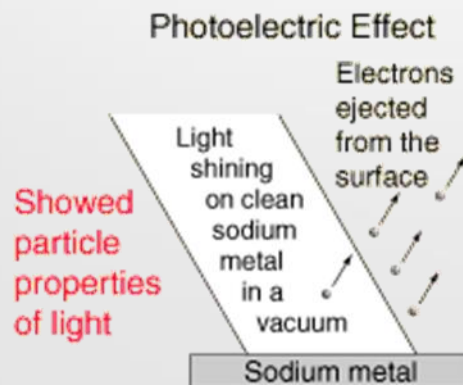
- In simple terms: When electrons reflect from a crystal, a diffraction pattern characteristic of wave interference is observed.



Science Photo Library/David Wisler,
Coloured by Dr. Jeremy Burgess

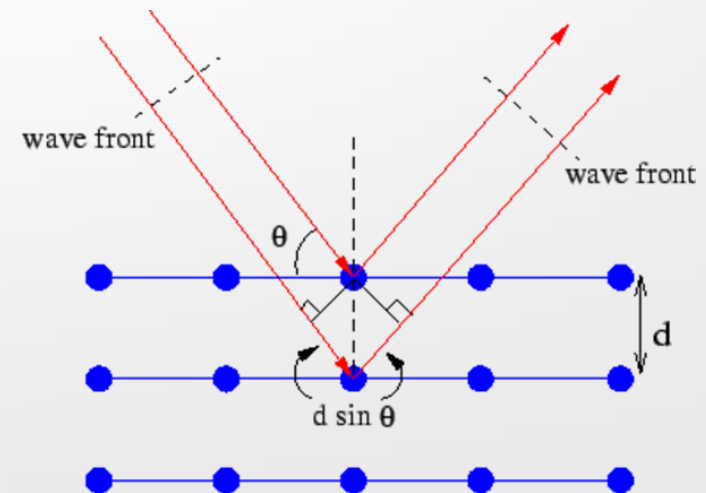
Davisson-Germer Experiment

- An electron beam was directed at a target which was a defined crystal structure made of nickel. The nickel could be rotated to observe angular dependence of the scattered electrons. The electrons could reflect off from different layers.
- Their electron detector was mounted on an arc so that it could be rotated to detect the intensity of the reflected electrons at different angles.
- The reflected electrons constructively and destructively interfered with each (a property of waves...)



- They found that at certain angles there was a peak in the intensity of the scattered electron beam. This peak indicated wave behaviour for electrons.
- When the extra distance travelled by the beam which travelled further into the nickel was an integer multiple of the wavelength ($n\lambda$) then the beams underwent constructive interference which increased the detected intensity.
- When the velocity was changed then the wavelength was changed ($\lambda = \frac{h}{mv}$) so now the

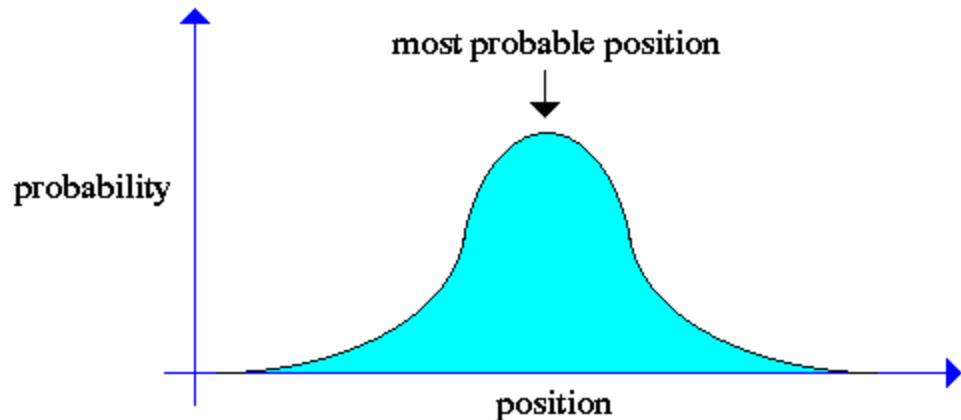
extra distance was no longer a integer multiple of wavelength and destructive interference occurred, which decreased the detected intensity.



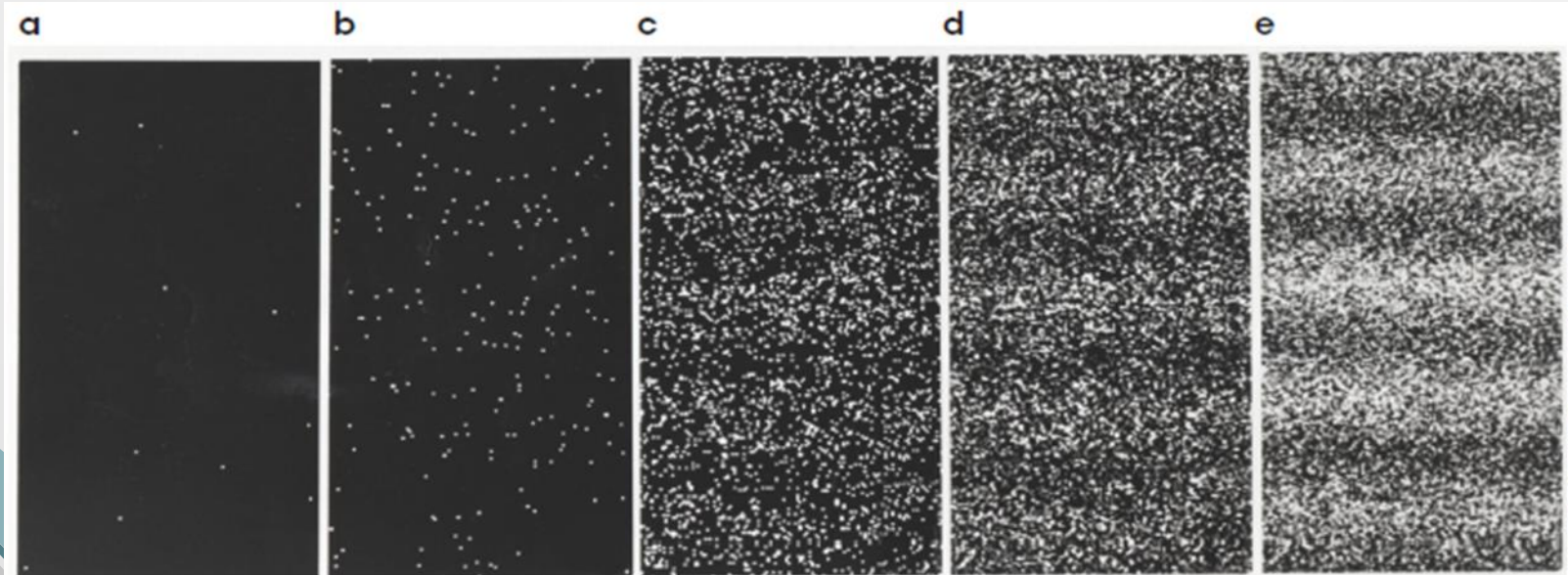
Particles as a wave

- Our understanding of particles is they have a defined location at a defined time.
- If particles have a wave nature then that means that particles do not have a defined location at a defined time.
- Instead particles have a probability distribution of where they could be.

Quantum Wave Function

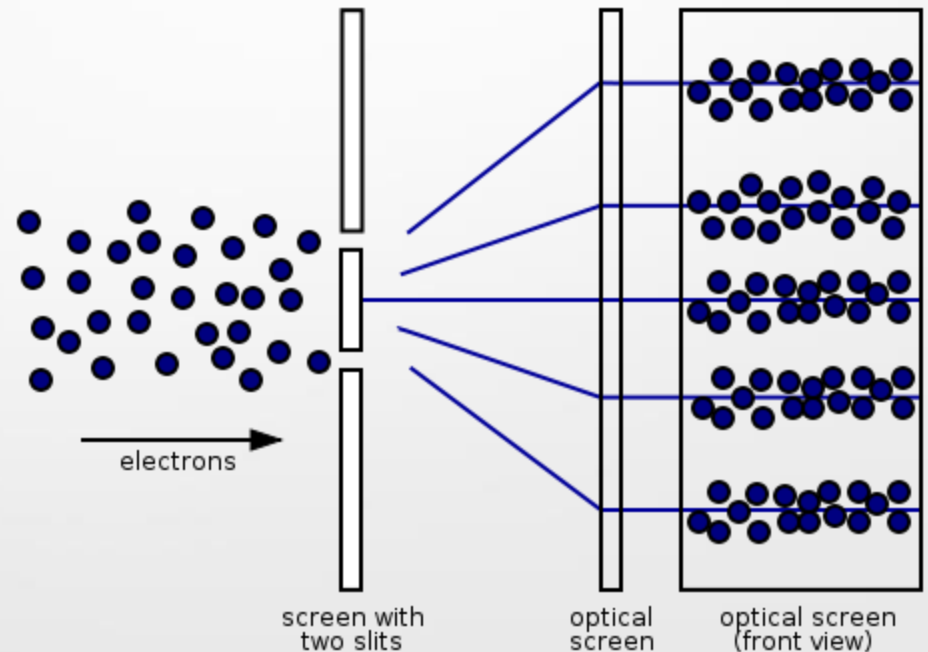


- Electron interference patterns build up in terms of probability distribution.
- There is a finite probability of the electron appearing at each point on the screen.
- After many single events the pattern emerges.
- What is interfering is the probability wave.



Young's double slit experiment with electrons

- In this version we replace the photons with electrons. The resulting pattern on the viewing screen is the same as observed with photons – a pattern of high intensity regions and low intensity regions.
 - Diffraction and interference occurs.
 - This pattern can only be explained if the electrons have a wave like nature



Question: What is the wavelength of an electron moving at a velocity of $3.00 \times 10^5 \text{ m s}^{-1}$?

$$\text{Answer: } \lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{9.11 \times 10^{-31} (3.00 \times 10^5)} = 2.43 \times 10^{-9} \text{ m}$$

Note: The size of the wavelength for the electron is much greater than the size of the electron. For everyday particles (like a basketball moving at a given velocity) the wavelength is smaller than the size of the object. This is why we do not see wave properties in everyday objects.

Question: What is the de Broglie wavelength of an electron moving with a kinetic energy of $2.00 \times 10^4 \text{ eV}$ (discounting relativistic effects)?

Answer: $E_k = \frac{1}{2}mv^2$ so $v = \sqrt{\left(\frac{2E_k}{m}\right)}$

$$E_k = (2 \times 10^4)(1.6 \times 10^{-19}) = 3.2 \times 10^{-15} \text{ J}$$

$$v = \frac{\sqrt{2(3.2 \times 10^{-15})}}{9.11 \times 10^{-31}} = 8.38 \times 10^7 \text{ m s}^{-1}$$

$$\lambda = \frac{h}{mv} = \frac{(6.63 \times 10^{-34})}{9.11 \times 10^{-31}(8.38 \times 10^7)} = 8.68 \times 10^{-12} \text{ m}$$

Question: What is the de Broglie wavelength of a bullet of mass 12.4 g moving at $2.51 \times 10^2 \text{ m s}^{-1}$?

$$\text{Answer: } \lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{(12.4 \times 10^{-3})(2.51 \times 10^2)} = 2.13 \times 10^{-34} \text{ m}$$